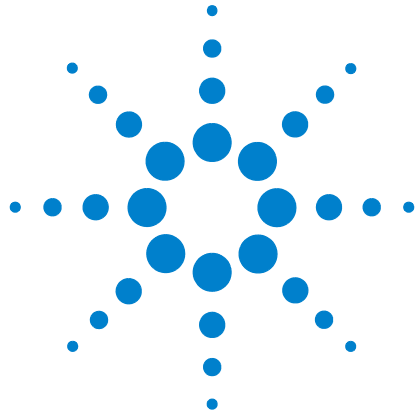




Agilent Magnetic Resonance Systems

Safety Guide



Agilent Technologies

Notices

© Agilent Technologies, Inc. 2013

No part of this manual may be reproduced in any form or by any means (including electronic storage and retrieval or translation into a foreign language) without prior agreement and written consent from Agilent Technologies, Inc. as governed by United States and international copyright laws.

Manual Part Number

G7446-35102

Edition

Rev.B, Jan. 2014

Revision Log

Rev.A, Nov. 2013	Initial release
Rev. B, Jan., 2014	Added Compliance Notice is Introduction chapter.

Printed in USA

Agilent Technologies, Inc.
5301 Stevens Creek Boulevard
Santa Clara, CA 95051 USA

Warranty

The material contained in this document is provided “as is,” and is subject to being changed, without notice, in future editions. Further, to the maximum extent permitted by applicable law, Agilent disclaims all warranties, either express or implied, with regard to this manual and any information contained herein, including but not limited to the implied warranties of merchantability and fitness for a particular purpose. Agilent shall not be liable for errors or for incidental or consequential damages in connection with the furnishing, use, or performance of this document or of any information contained herein. Should Agilent and the user have a separate written agreement with warranty terms covering the material in this document that conflict with these terms, the warranty terms in the separate agreement shall control.

Technology Licenses

The hardware and/or software described in this document are furnished under a license and may be used or copied only in accordance with the terms of such license.

Restricted Rights Legend

If software is for use in the performance of a U.S. Government prime contract or sub-contract, Software is delivered and licensed as “Commercial computer software” as defined in DFAR 252.227-7014 (June 1995), or as a “commercial item” as defined in FAR 2.101(a) or as “Restricted computer software” as defined in FAR 52.227-19 (June 1987) or any equivalent agency regulation or contract clause. Use, duplication or disclosure of Software is subject to Agilent Technologies’ standard commercial license terms, and non-DOD Departments and Agencies of the U.S. Government will receive no greater than Restricted Rights as defined in FAR 52.227-19(c)(1-2) (June 1987). U.S. Government users will receive no greater than Limited Rights as defined in FAR 52.227-14

(June 1987) or DFAR 252.227-7015 (b)(2) (November 1995), as applicable in any technical data.

Safety Notices

CAUTION

A **CAUTION** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a **CAUTION** notice until the indicated conditions are fully understood and met.

WARNING

A **WARNING** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in personal injury or death. Do not proceed beyond a **WARNING** notice until the indicated conditions are fully understood and met.

Contents

1 Introduction

Warning Notices	6
Caution Notices	8
Compliance Notice	9
South Korean Class A EMC declaration	9
Marking	9

2 Magnet Safety

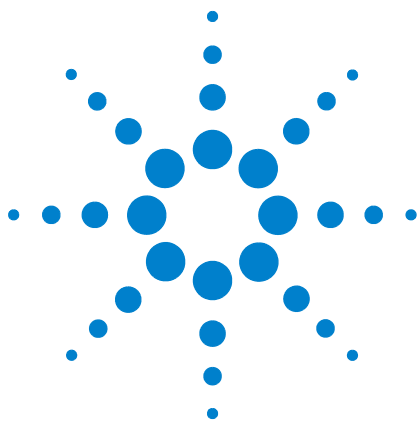
General Preventative Safety	11
Magnetic Field Hazards	12
Magnetic Field Exclusion Zone (5-gauss line)	13
Consideration of Magnetic Field on Equipment	14
Occupational Exposure to Magnetic Fields	14

3 General Lab Safety

Compressed Gas Cylinders	15
Adequate Ventilation During Magnet Installation	16
Oxygen Monitor	16
Electrical Safety	17
Electrostatic Discharges	17
Radio-Frequency Emission Regulations	18
Safe Access When Working at Height	18
Considerations for the Outbreak of Fire	18

4 Cryogen Safety

Nature of Liquid Cryogens	21
Hazards Presented by Cryogens	21
Asphyxiation hazards	22
Cryogenic burns	22
Fire hazards	22
Handling Cryogens (Liquid Helium and Liquid Nitrogen)	23



1

Introduction

Warning Notices	6
Caution Notices	8
Compliance Notice	9

The site for an NMR spectrometer system with a superconducting magnet must meet national, state, and local safety regulations. For NMR spectrometer systems, site safety issues primarily relate to magnetic fringe fields and handling liquid cryogenes.

Special considerations and precautions in this document appear in the form of **WARNINGS**, **CAUTIONS**, and **NOTES** as described below. Operate the equipment in accordance with this instruction manual and any additional information provided by Agilent Technologies, Inc.

Failure to comply with these warnings or with specific warnings elsewhere in Agilent manuals, violates safety standards of design, manufacture, and intended use of the instrument. Agilent assumes no liability for customer failure to comply with these precautions.

Always refer to your local laws and ordinances regarding requirements for design and posting of safety signs.



WARNING

A **WARNING** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in personal injury or death. Do not proceed beyond a **WARNING** notice until the indicated conditions are fully understood and met.

CAUTION

A **CAUTION** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a **CAUTION** notice until the indicated conditions are fully understood and met.

NOTE

A **NOTE** provides important information on how to obtain optimal performance from the instrument.

Warning Notices

WARNING

Persons with implanted or attached medical devices such as pacemakers and prosthetic parts must remain outside the 5-gauss perimeter from the centerline of the magnet.

WARNING

The strong magnetic field surrounding the magnet attracts objects containing steel, iron, or other ferromagnetic materials, which includes most ordinary tools, electronic equipment, compressed gas cylinders, steel chairs, and steel carts. Unless restrained, such objects can suddenly fly towards the magnet, causing possible personal injury and extensive damage to the probe, Dewar, and superconducting solenoid. The greater the mass of the object, the more the magnet attracts the object.

WARNING

SHOCK HAZARD

Only qualified maintenance personnel shall remove equipment covers or make internal adjustments.

Dangerous high voltages that can kill or injure exist inside the instrument. Before working inside a cabinet, turn off the main system power switch located on the back of the console.

WARNING

Do not substitute parts or modify the instrument.

Any unauthorized modification could injure personnel or damage equipment and potentially terminate the warranty agreements and/or service contract. Written authorization approved by an Agilent, Inc. product manager is required to implement any changes to the hardware of an Agilent NMR spectrometer. Maintain safety features by referring system service to an Agilent service office.

WARNING**FIRE and EXPLOSION HAZARD**

Do not operate in the presence of flammable gases or fumes.

Operation with flammable gases or fumes present creates the risk of injury or death from toxic fumes, explosion, or fire.

WARNING

Leave area immediately in the event of a magnet quench.

A quench causes a sudden release of helium or nitrogen gases that can rapidly displace oxygen in an enclosed space creating a possibility of asphyxiation. Do not return until the oxygen level returns to normal.

WARNING**BURN HAZARD**

Avoid direct contact with cryogenic liquids or cold gas, including equipment that has been in direct contact with them.

The extreme cold temperatures cause burns to the skin. Never place any part of your body over the helium and nitrogen exit tubes on top of the magnet. In case of cold burns, seek immediate medical attention, especially if the skin is blistered or the eyes are affected.

WARNING**EYE HAZARD**

Do not look down the upper barrel.

Unless the probe is removed from the magnet, never look down the upper barrel. The sample tube could injure you as it ejects pneumatically from the probe.

WARNING

EXPLOSION HAZARD

Do not exceed the boiling or freezing point of a sample during variable temperature experiments.

A sample tube subjected to a change in temperature can build up excessive pressure, which can break the sample tube glass and cause injury by flying glass and toxic materials. To avoid this hazard, establish the freezing and boiling point of a sample before doing a variable temperature experiment.

Caution Notices

CAUTION

MAGNETIC MEDIA DAMAGE

Keep magnetic media, automatic teller machine (ATM) and credit cards, and watches outside the 5-gauss perimeter of the magnet.

The strong magnetic field surrounding a superconducting magnet can erase magnetic media such as floppy disks and tapes. The field can also damage the strip of magnetic media found on credit cards, ATM cards, and similar plastic cards. Many wrist and pocket watches are also susceptible to damage from intense magnetism. Refer to the manuals supplied with the magnet for the size of a typical 5-gauss stray field. This gauss level should be checked after the magnet is installed.

CAUTION

COMPUTER DAMAGE AND DATA LOSS

Keep computers beyond the 5-gauss perimeter of the magnet.

Avoid equipment damage or data loss by keeping PCs well away from the magnet. Generally, keep the PC beyond the 5-gauss perimeter of the magnet. Refer to the stray field data provided in the magnet manual.

CAUTION

ESD DAMAGE

Take electrostatic discharge (ESD) precautions to avoid damage to sensitive electronic components.

Wear grounded antistatic wristband or equivalent before touching any parts inside the doors and covers of the spectrometer system. Also, take ESD precautions when working near the exposed cable connectors.

Compliance Notice

Declarations of Conformity are available at:

<http://regulations.corporate.agilent.com/DoC/search.htm>.

South Korean Class A EMC declaration

This equipment is Class A suitable for professional use and is for use in electromagnetic environments outside of the home.

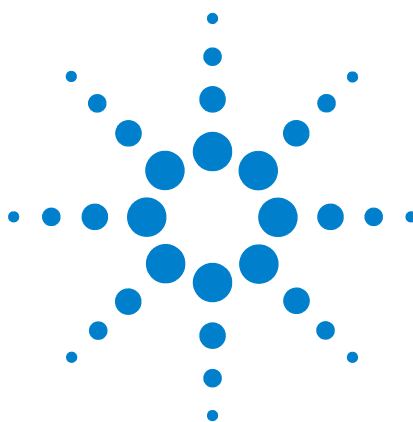
A급 기기

(업무용 방송통신기자재)

이 기기는 업무용(A급) 전자파적합기기로서 판매자 또는 사용자는 이 점을 주의하시기 바라며, 가정외의 지역에서 사용하는 것을 목적으로 합니다.

CE Marking

Marking by the  symbol indicates compliance of this device to the EMC (electromagnetic compatibility) and LV (low voltage) directives of the European Community.



2 Magnet Safety

General Preventative Safety	11
Magnetic Field Hazards	12
Magnetic Field Exclusion Zone (5-gauss line)	13
Consideration of Magnetic Field on Equipment	14
Occupational Exposure to Magnetic Fields	14

General Preventative Safety

Follow these safety practices to ensure safe equipment operation:

- Perform periodic leak checks on all supply lines and pneumatic plumbing.
- Do not allow gas lines to become kinked or punctured.
- Place all supply lines away from foot traffic and extreme heat or cold.

WARNING

Keep cryostat exhaust ports free from blockages such as ice.

Ensure all safety devices (relief valves and drop-off plate) are maintained in working order.

Do NOT tamper with any vacuum seals or attempt to unscrew vacuum feed-through connections.

WARNING

Do not remove the pressure relief valves on the vent tubes.

The relief valves prevent air from entering the nitrogen and helium vent tubes. Air that enters the magnet can freeze, causing blockage of the vent tubes resulting in dangerous build-up of pressure in the vessel. Except when transferring nitrogen or helium, be certain that the relief valves are secured on the vent tubes.

CAUTION

Check helium and nitrogen gas flow meters daily, if supplied with the system.

Record the readings to establish the operating level. The readings will vary somewhat because of changes in barometric pressure from weather fronts. If the readings for either gas should change abruptly, contact qualified maintenance personnel. Failure to correct the cause of abnormal readings could result in extensive equipment damage.



Magnetic Field Hazards

The strong magnetic field creates hazards because it strongly attracts ferromagnetic metal alloys. Ferromagnetic metal alloys contain iron, nickel, or cobalt. They are used in most types of tools and equipment and in some surgical implants.

Once the magnet is energized, the magnetic field is always present, even when all power to the system is turned off. This magnetic field extends above, below and to the sides of the magnet. It extends through doors, walls ceilings, and floors.

The strength of the magnetic field increases as the magnet is approached. Objects and people at risk from the magnetic field are considered safe outside the 5 gauss line.

The strong magnetic field can cause the following hazards:

- Projectiles
- Displacement/stoppage of surgical implants

WARNING

Use only nonmagnetic tools near the magnet. Use great care if nonmagnetic tools are unavailable and you must use steel tools near the magnet.

If a loose metal object (such as a wrench or pen) gets too close to the magnet, it can become a projectile as the force of the magnetic field pulls the object towards the magnetic center. This metal projectile can seriously injure anyone standing between it and the magnet.

Magnetic fields are particularly high at the ends of the bore tube (that is, directly underneath and on top of vertical magnets).

In the rare event of a magnet quench, it is possible for the magnetic field to bloom momentarily beyond its normal limit, attracting ferromagnetic objects towards the magnet.

WARNING

The interaction between the magnetic field and surgical implants can cause a life-threatening situation.

Ferromagnetic metals are sometimes used in surgical implants and prosthetic devices. The magnetic field can twist metallic implants out of place, causing tissue damage and pain to the person in whom they are implanted.

Some cardiac pacemakers, biostimulators, and neurostimulators are mechanically activated and can stop in the presence of the magnetic field.

Use only nonmagnetic tools near the magnet. Even some stainless steel tools can be magnetic. Position any appliances that require regular servicing, such as air conditioning units

or radiators, far enough away to enable safe use of steel tools. Permit only nonmagnetic fire extinguishers within the magnet room.

NOTE

A magnet that has been de-energized but remains at liquid helium temperatures still exhibits a small residual magnetic field. Make measurements of this field, and restrict public access as per national and state requirements.

Magnetic Field Exclusion Zone (5-gauss line)

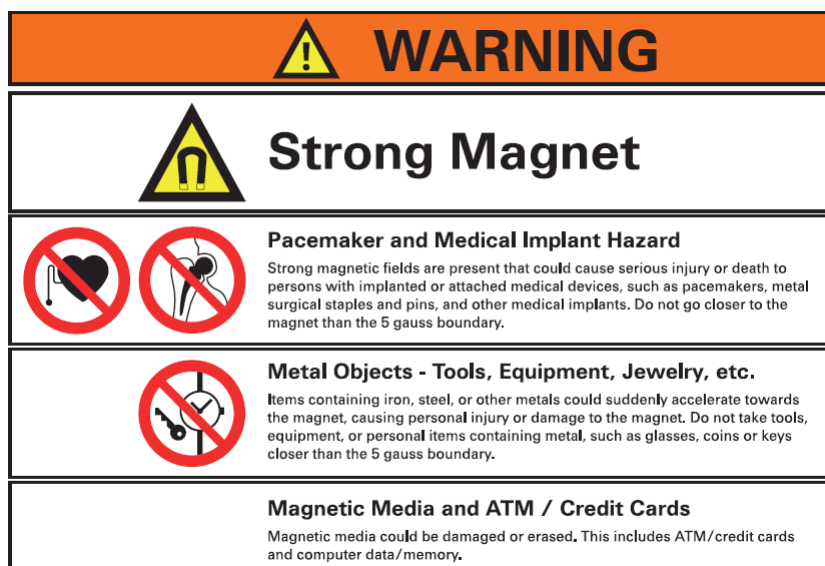
WARNING

Maintain a safety zone around the magnet at the 5G (0.5 mT) line. Keep the following outside this line:

- > People with heart pacemakers or metallic body implants.
- > Gas cylinders.
- > Ferromagnetic objects.
- > Electronic components containing transformers, which can contain iron.

The exclusion zone includes the area (rooms and hallways for example) inside the 5-gauss line of the magnet. Prevent individuals with cardiac or other mechanically active implants from entering this area.

Enforce the exclusion zone with a combination of warning signs and physical barriers. The following figure shows the warning sign recommended by Agilent Technologies.



Consideration of Magnetic Field on Equipment

Table 1 lists some magnetically sensitive devices and guidelines for the acceptable fringe field (in gauss) within which they can effectively operate.

Table 1 Equipment affected by magnetic field

Magnetic field	Objects affected by the magnetic field
10 gauss line or closer	Ferromagnetic implant, and unrestrained ferromagnetic objects, such as tools, keys, electronic equipment, analog watches, magnetic data storage media, and credit cards.
5 to 10 gauss	Cardiac pacemakers and electronic equipment, such as shielded CRT monitors, computers, shielded image intensifiers, and shielded photomultiplier tubes.
1 to 5 gauss	Very sensitive electronic equipment, such as unshielded image intensifiers, photomultiplier tubes, linear accelerators, electron microscopes, and CRT monitors.

Occupational Exposure to Magnetic Fields

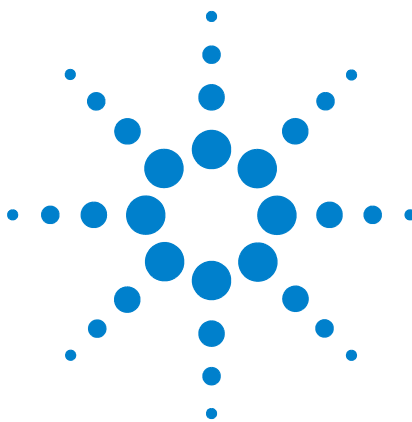
Consult national and state enforcement and advisory bodies to ensure that any exposure limits are not infringed for (nonpatient) occupational workers.

NOTE

National and state legislation & requirements differ around the world. In the USA, follow the requirements and guidance issued by the Food & Drug Administration (FDA). In Europe, European Commission (EC) Directive 2004/40/EC applies; follow the guidance issued by the International Commission on Non Ionizing Radiation Protection (ICNIRP).

NOTE

Before 1st April 2005, the advisory body for the United Kingdom was the National Radiological Protection Board (NRPB) - a member of ICNIRP. From 1st April 2005 the NRPB merged with the United Kingdom Health Protection Agency (HPA), and became its Radiation Protection Division.



3 General Lab Safety

Compressed Gas Cylinders	15
Adequate Ventilation During Magnet Installation	16
Oxygen Monitor	16
Electrical Safety	17
Electrostatic Discharges	17
Radio-Frequency Emission Regulations	18
Safe Access When Working at Height	18
Considerations for the Outbreak of Fire	18

Compressed Gas Cylinders

- Store and handle compressed gases carefully and in strict adherence to safety codes.
- Secure cylinders to an immovable structure or wall.
- Store and move cylinders in an upright, vertical position. Before transport, remove regulators and install cylinder caps.
- Store cylinders in a well-ventilated area away from heat, direct sunshine, freezing temperatures, and ignition sources.
- Mark cylinders clearly so there is no doubt as to their contents.
- Use only approved regulators and connections.



Adequate Ventilation During Magnet Installation

During the installation and cooling of superconducting magnets, large volumes of nitrogen or helium gases are sometimes generated. Although these gases are inert, if generated in large enough quantities, they can create dangerous circumstances if they displace the oxygen in the room. Refer to the *Site Planning Guide* or the Magnet User Guide supplied with your magnet for room ventilation requirements.

WARNING

Reduce the risk of asphyxiation by maintaining adequate means of removing evolved gases from the magnet out of the magnet room.

See “[Handling Cryogens \(Liquid Helium and Liquid Nitrogen\)](#)” on page 23.

Oil mist from rotary vane vacuum pumps is a potential health hazard: vent the exhaust from pumps to the outdoors. Make sure any flexible hoses used for this purpose are kink-resistant and have a push-on engagement with the exhaust port of the vacuum pump. (Do not clamp hoses onto pumps.)

Oxygen Monitor

NOTE

An oxygen monitor is recommended for the magnet room. See the *Site Planning Guide* for recommendations. Make sure the audio/visual alarm connected to an oxygen monitor is visible/audible outside the magnet room as well as inside it.

The oxygen monitor sounds an alarm when the oxygen level in the room rises above or falls below a safe level. This alarm happens if the ventilation system malfunctions, either during a quench or during a normal boil-off, and a large amount of helium gas is released into the room.

Electrical Safety

- Disconnect the instrument from all power sources before removing protective panels to avoid exposure to potentially dangerous voltages.
- When it is necessary to use a non-original power cord plug, ensure the replacement cord adheres to the color-coding and polarity described in the manual and all local building safety codes or local electrical legislation.
- Replace blown fuses with fuses of the size and rating stipulated on the fuse panel or in the manual.
- Replace faulty or frayed power cords immediately with the same type and rating.
- Make sure that voltage sources and line voltage match the value for which the instrument is wired.

Always observe the following precautions and safe practices:

WARNING

Electric shock hazard. Isolate or disconnect equipment from mains supply before opening the unit for any reason.

WARNING

For continued protection against risk of fire or electric shock, always replace fuses with the same type, rating, and UL listing.

Electrostatic Discharges

Electrostatic discharge (ESD) can damage electronic components.

Whenever a printed circuit board must be touched or handled, the person should wear grounded wrist straps.

CAUTION

Many components in the system contain highly sensitive electronic devices that must be protected from electrostatic discharges by proper floor coverings and grounding practices.

As with any system based on integrated circuits, the system is susceptible to static spikes, both- those generated on the power line and those generated in the lab area, must be suppressed.

Radio-Frequency Emission Regulations

The covers on the instrument form a barrier to radio-frequency (rf) energy. Removing any of the covers or modifying the instrument may lead to increased susceptibility to rf interference within the instrument and may increase the rf energy transmitted by the instrument in violation of regulations covering rf emissions. It is the operator's responsibility to maintain the instrument in a condition that does not violate rf emission requirements.

Safe Access When Working at Height

Safe access to the top of magnets is important for routine topping up of cryogenics; the frequency of doing so varies from magnet to magnet, but can be needed as often as weekly. Safe access to the top of magnets is also essential for installation work. Normal ladders and stepladders are not best suited to this task: They often fail to offer a secure platform when working on top of magnets and performing tasks that require the use of both hands.

Considerations for the Outbreak of Fire

WARNING

Although large fires are unlikely, always evacuate all staff before a fire becomes fully established, following standard fire procedures or evacuation plans.

Specific precautions regarding fire safety vary depending on legislation and regulations in the country where the magnet is installed. The considerations discussed in the following sections are intended to help customers review their site fire precautions following the installation of a superconducting magnet. **These considerations are advisory in nature.**

There are two types of fire that can affect a magnet room – small fires and large fires. In this document “small fires” are defined as fires of a size that personnel can choose to fight with a first aid fire extinguisher. “Large fires” are defined here as major fires that only professional fire fighters are qualified to extinguish.

Small fires. Most premises already have fire and emergency procedures in place, in accordance with current safety legislation. Make sure any fire extinguisher provided in a magnet room, or taken into a magnet room in response to a fire, **is of nonferromagnetic construction**. Otherwise, it can become attracted to the magnet, causing personal injury, and becoming useless for fire fighting purposes.

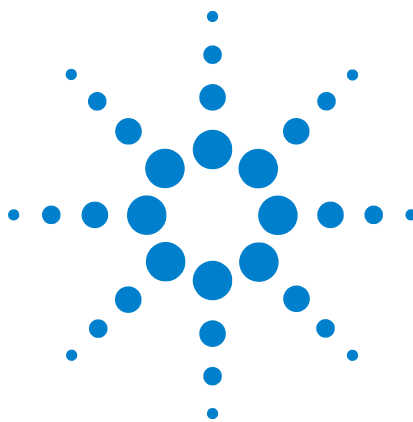
Large fires. Most premises already have fire and emergency procedures in place, in accordance with current safety legislation. **For maximum safety, invite the Fire Department (or Fire Brigade) for a liaison visit to inspect the building layout and the hazards associated with superconducting magnets.** It is important for the Fire Department (or Brigade) to understand – before attending a major fire at a magnet lab – the hazards *in addition to* flames and smoke that these installations represent. The hazards to discuss with professional fire fighters include the following:

Magnetic attraction of ferrous equipment. Assuming the magnetic field warning signs are still intact when fire fighters, most firemen do not understand the strength of magnetic attraction that superconducting magnets generate. These magnetic fields can attract fire fighting equipment and endanger the firefighters themselves.

Quenches. Quench ducts can fail in a large fire or be broken by falling debris, liberating quench gases into the magnet room, creating a risk of asphyxiation for any fire fighters present. Magnet rooms with several vertical magnets, which do not possess dedicated quench ducts, can be faced with more than one magnet quenching simultaneously, venting large quantities of nitrogen or helium gas into the magnet room. Because these gases are inert, as few as three breaths of these gases can cause asphyxiation. Magnet rooms of very large volumes can represent a lower risk than magnet rooms of small volumes. However, always advise fire fighters of the asphyxiation hazard, and suggest they wear breathing apparatus when fighting a fire in a magnet room.

Explosions. In a disaster scenario, magnets very close to or enveloped in flames get very hot, causing them to quench and rapidly release inert gases, possibly faster than their pipes and fittings normally permit. In the worse case, a magnet can become over pressurized and explode. If magnets become physically hot in a fire, Treat them in a similar way to hot compressed gas cylinders: hosed down and kept cool.

Fire Doors and Exits. Make sure fire doors and exits from magnet rooms open outwards. In a disaster scenario, if large quantities of gas are vented into the magnet room, the pressure in the room can rise, making inward-opening doors difficult to open. This situation can make access difficult for fire fighters, or potentially fatal for anyone still inside the magnet room.



4 Cryogen Safety

Nature of Liquid Cryogens 21

Hazards Presented by Cryogens 21

Handling Cryogens (Liquid Helium and Liquid Nitrogen) 23

Nature of Liquid Cryogens

A superconducting magnet uses two types of cryogenics: liquid helium and liquid nitrogen. Both are colorless, odorless, nonflammable, and nontoxic.

- Helium is a naturally occurring, inert gas that becomes a liquid at approximately 4K. In order to remain in a superconducting state, the magnet is immersed in liquid helium.
- Nitrogen is a naturally occurring gas that becomes liquid at approximately 77K. It is used to cool the shield that surrounds the liquid helium reservoir.

During normal operation, liquid cryogenics evaporate and require replenishment on a regular basis.

CAUTION

Cryogenics are delivered to the site in Dewars; these Dewars, including any carriers, must be nonmagnetic.

Hazards Presented by Cryogenics

Helium and nitrogen in both their liquid and gaseous forms, present the following hazards:

- A large quantity of gas released in an area that is not well ventilated can cause asphyxiation.
- Liquids or cold gases can cause cryogenic burns.
- The filling procedure produces an oxygen-rich liquid, which is a fire hazard when it drops on a combustible material.



Asphyxiation hazards

WARNING

Helium or nitrogen gas released in an area that is not well ventilated can displace air and reduce the oxygen content to an unsafe level, leading to asphyxiation.

Also note that, when cold, gaseous helium remains lighter than air, but gaseous nitrogen is heavier than air. The greatest risk of asphyxiation is during commissioning of the magnet. However, after the magnet is in service, a risk of asphyxiation can still occur during a “quench”.

A “quench” occurs when the magnet loses its superconductivity, and warms the liquid helium, causing it to turn to gas. This large amount of gas escapes through the magnet venting. The following section describes the safety recommendations that prevent the possibility of asphyxiation after a magnet quench.

Cryogenic burns

WARNING

Touching the fill line or any other piece of equipment during a transfer can freeze the skin or cause it to stick to the surface of the equipment.

The extremely low temperature of the liquids and their cold vapors can cause severe frostbite, or a cryogenic burn. These low temperatures are mainly a danger during the filling process, when the fill line and other equipment freezes.

To prevent cryogenic burns, wear safety footwear, protective gloves, and a face shield. Long-sleeved clothing and long trousers or pants are also recommended.

Fire hazards

WARNING

Liquid oxygen can spontaneously ignite upon contact with oils or greases.

During the filling process, atmospheric air condenses on the fill line or pipes, leaving an oxygen-rich liquid, or even liquid oxygen. If this drips onto a combustible material, like oil or grease, it can cause a fire.

Review the following recommendations:

- Exclude all sources of ignition from magnet rooms.
- Keep all tools and equipment used for transferring cryogenic liquids clean and free from oils and grease.
- Exclude all sources of ignition from the magnet room, which is also designated as a No Smoking zone.

Handling Cryogenics (Liquid Helium and Liquid Nitrogen)

WARNING

Allow only personnel with formal training to perform routine maintenance tasks involving the transfer of cryogenic liquids.

Wear safety goggles, loose fitting nonabsorbent gloves, long trousers, a long-sleeved shirt, and nonferrous safety footwear.

Ensure adequate ventilation and oxygen monitoring of the area consistent with the volume of cryogenics being used. If the oxygen level in the room falls below 19.5%, do not allow personnel to enter the area without self-contained breathing apparatus or airline respirators.

Do not smoke. Ensure that other sources of ignition are not present. Liquid oxygen can condense on cold surfaces and cause a fire hazard.

Never measure the helium level by “dipping” or “thumping” the helium can; always use the integral level probes in order to gauge the volume of liquid remaining.



© Agilent Technologies, Inc.

Printed in USA, Rev.B, Jan. 2014



G7446-35102